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Optical isolator core

Abstract

In some implementations, an optical isolator core includes a Faraday rotator and a plurality of birefringent crystal plates. The plurality of birefringent crystal plates may include a first birefringent crystal plate to separate input light into light having a first polarization and light having a second polarization, and a second birefringent crystal plate to combine the light having the first polarization and the light having the second polarization in output light that is laterally displaced by the single stage optical isolator. The Faraday rotator may be provided between the first birefringent crystal plate and the second birefringent crystal plate. In some implementations, the plurality of birefringent crystal plates further include a third birefringent crystal plate provided between the Faraday rotator and the second birefringent crystal plate. Additionally, or alternatively, the optical isolator core may further include a half-wave plate arranged between the Faraday rotator and the first birefringent crystal plate.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5066092	12/1990	Droegemueller	385/33	G02F 1/093
5493440	12/1995	Souda et al.	N/A	N/A
5835270	12/1997	Urino et al.	N/A	N/A
5841573	12/1997	Kim	385/24	G02B 6/2746
6088153	12/1999	Anthon et al.	N/A	N/A
6430323	12/2001	Kokkelink et al.	N/A	N/A
12034266	12/2023	Chen et al.	N/A	N/A
2001/0036330	12/2000	Delisle	385/11	G02B 27/283
2001/0053022	12/2000	Tai	359/484.03	H01S 3/06754
2002/0089745	12/2001	Huang et al.	N/A	N/A
2003/0228082	12/2002	Chang	385/11	G02B 6/2713
2005/0280888	12/2004	Pan et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1317897	12/2000	CN	N/A
2499864	12/2001	CN	N/A
1365011	12/2001	CN	N/A
1766725	12/2005	CN	N/A
201072472	12/2007	CN	N/A
202794598	12/2012	CN	N/A

103576346	12/2013	CN	N/A
108020924	12/2017	CN	N/A
108700791	12/2017	CN	N/A
210109495	12/2019	CN	N/A
0939465	12/1998	EP	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for Application No. PCT/CN2020/088952, mailed on Jan. 27, 2021, 9 pages. cited by applicant

International Search Report and Written Opinion for Application No. PCT/CN2020/119360, mailed on Jan. 28, 2021, 10 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. patent application Ser. No. 17/164,191, filed Feb. 1, 2021 (now U.S. patent application Ser. No. 12/034,266), which claims priority to Patent Cooperation Treaty (PCT) Application No. PCT/CN2020/088952, entitled “OPTICAL ISOLATOR CORE IN BETWEEN FIBER AND COLLIMATOR LENS,” filed on May 7, 2020, and PCT Application No. PCT/CN2020/119360, entitled “OPTICAL ISOLATOR CORE,” filed on Sep. 30, 2020, the contents of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

(1) The present disclosure relates generally to optical systems and to optical isolator core designs to provide a lateral displacement walk-off for light received at the optical isolator core.

BACKGROUND

(2) An optical isolator (sometimes referred to as an optical diode or a Faraday isolator, among other examples) is an optical device that transmits light in a certain direction while blocking light in an opposite direction. Optical isolators are typically used to prevent unwanted feedback from outside of an optical oscillator, such as a laser cavity. For example, in some cases (e.g., at high powers), optical feedback can damage or disrupt the operation of a laser system. To reduce the optical feedback, an optical isolator can be inserted into the laser system. The optical isolator may be a passive unidirectional, nonreciprocal device that utilizes the phenomenon of magneto-optic rotation to isolate a source and protect a laser oscillator from reflections.

SUMMARY

(3) In some implementations, an optical assembly includes a first optical fiber and a second optical fiber; a pump laser to provide pump light having a first wavelength, the pump light coupling into the first optical fiber through a first collimator lens, a wavelength-division multiplexing filter, a second collimator lens, and a compensator arranged between the first optical fiber and the second collimator lens; and an isolator arranged between the second optical fiber and the second collimator lens to transmit signal light having a second wavelength, wherein the isolator includes a Faraday rotator and a plurality of birefringent crystal plates to laterally displace the signal light.

(4) In some implementations, a single stage optical isolator includes a Faraday rotator; and a plurality of birefringent crystal plates, wherein the plurality of birefringent crystal plates include a first birefringent crystal plate to separate input light into light having a first polarization and light

having a second polarization, and a second birefringent crystal plate to combine the light having the first polarization and the light having the second polarization in output light that is laterally displaced by the single stage optical isolator, wherein the Faraday rotator is provided between the first birefringent crystal plate and the second birefringent crystal plate.

(5) In some implementations, a dual stage optical isolator includes a first isolator core comprising a first set of layers including a first half-wave plate and a first Faraday rotator arranged between a first pair of birefringent crystal plates; and a second isolator core comprising a second set of layers including a second half-wave plate and a second Faraday rotator arranged between a second pair of birefringent crystal plates, wherein the first set of layers and the second set of layers have identical thicknesses and respective magnetic orientations to cancel a material tolerance and an assembly tolerance with respect to one another.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1A-1E illustrate one or more example implementations of a single stage isolator.

(2) FIGS. 2A-2C illustrate one or more example implementations of a dual stage isolator including two isolator cores that are arranged in parallel.

(3) FIGS. 3A-3C illustrate one or more example implementations of a dual stage isolator including two isolator cores that are arranged in series.

DETAILED DESCRIPTION

(4) The following detailed description of example implementations refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

(5) An optical isolator is a device that is generally designed to allow light to travel in a forward direction while blocking reflections that would travel in a backward direction. Optical isolators are important in many optical systems and applications. For example, a single-frequency semiconductor laser is very susceptible to external optical feedback. Even a very low level of optical reflection from an external optical circuit (e.g., on the order of -50 dB) is sufficient to cause a significant increase in laser phase noise, intensity noise, and/or wavelength instability, among other examples. Accordingly, an optical isolator is often provided at the output of each laser diode in applications that require low optical noise and stable single optical frequency. Another example is in an optical amplifier where unidirectional optical amplification is desired. In this case, the bidirectional nature of optical amplification of the optical gain medium would cause self-oscillation if the external optical reflection from, for example, connectors and other optical components is strong enough.

(6) A traditional optical isolator is based on a Faraday rotator provided between two polarizers (e.g., an input polarizer and an output polarizer). In this configuration, input light received at the optical isolator passes through the input polarizer, which has an optical axis that matches a polarization orientation of the input light. The Faraday rotator then rotates the polarization of the input light signal by 45 degrees in a clockwise direction. The optical axis of the output polarizer is oriented 45 degrees with respect to the input polarizer, which allows the optical signal to pass through with little attenuation. If there is a reflection from one or more components downstream from the output polarizer, the reflected light has to pass through the output polarizer and the Faraday rotator. Because the Faraday rotator is a nonreciprocal device, the polarization state of the reflected light will rotate an additional 45 degrees in the same direction as the input light, thus becoming orthogonal to the optical axis of the input polarizer. The input polarizer therefore effectively blocks the reflected light and ensures unidirectional light transmission through the optical isolator.

(7) Accordingly, in a traditional optical isolator design, an optical core deflects a collimated light beam with a small angle, which may be used in a collimated light beam application. For example, one collimated light beam application may involve integrating an isolator core with a wavelength division multiplexer (WDM) filter and pump chip in an optical assembly that combines light beams having different wavelengths (e.g., a pump combiner). For example, the optical assembly may include an input leg that receives an input signal at a first wavelength (e.g., 1550 nanometers (nm)) and an output leg in which the input signal at the first wavelength is combined with pump light having a second wavelength (e.g., 980 nm). The typical approach in such collimated light beam applications is to place the optical isolator in an area between the WDM filter and a collimator lens, which has the most physical space to integrate the optical isolator within the device. However, if the optical isolator is placed in the area between the WDM filter and the collimator lens, the optical isolator may block the pump light that is traveling in a reverse direction relative to the input light (e.g., because optical isolators typically include material, such as garnet, that has a high absorption for 980 nm pump light).

(8) Some implementations described herein provide an isolator core and/or an optical system (e.g., an optical assembly and/or an optical device) that includes an isolator core designed to provide a lateral displacement walk-off. For example, rather than deflecting light in an input beam received at the isolator core, the isolator core may include a Faraday rotator, a plurality of birefringent crystal plates, and/or a half-wave plate (HWP) arranged to spatially separate and/or combine orthogonal polarization components of the input beam with low polarization mode dispersion (PMD), polarization dependent loss (PDL), and/or insertion loss (IL). Furthermore, the isolator core may have a design that prevents backward travelling light entering an output side of the optical isolator from coupling into an input side of the optical isolator, which may achieve optical isolation. In some implementations, the isolator core may be positioned in an optical assembly between an optical fiber and a collimator lens to enable a same side input/output fiber (i.e., same side of an assembly), an integrated 980 nm/1550 nm WDM function, self-compensated PMD, and/or self-compensated PDL. In some implementations, the isolator core may have a design that conserves space by utilizing space between the optical fiber and the collimator lens. Furthermore, the isolator core design may provide flexibility to place the isolator core in an input leg of an optical assembly, an output leg of an optical assembly, or both the input and output legs of an optical assembly. For example, in a forward pump erbium doped fiber amplifier (EDFA) where 980 nm pump light is coupling through a WDM filter into the output leg, the isolator core may be arranged in the input leg (e.g., because the isolator core may otherwise block the 980 nm pump light). In another example, in a reverse pump EDFA where 980 nm pump light couples through a WDM filter into the input leg, the isolator core may be provided in the output leg. In some implementations, the optical isolator design may enable an input fiber and an output fiber to be on the same side of an assembly, which may enable integration with a 980 nm pump laser chip and/or a monitor photodiode.

(9) FIGS. 1A-1E illustrate one or more example implementations **100** of a single stage isolator **110**. As shown in FIGS. 1A-1C, for a forward pumped EDFA, the single stage isolator **110**, which may be referred to herein as isolator core **110**, may be arranged in an optical assembly to laterally displace an input light beam received at the isolator core **110**. For example, as shown, the isolator **110** may be used in an optical assembly, such as a pump combiner, that includes or interfaces with a dual fiber pigtail **120** that includes an input optical fiber in an input leg of the optical assembly and an output optical fiber in an output leg of the optical assembly. As shown in FIGS. 1A-1C, the single stage isolator core **110** may be positioned between the dual fiber pigtail **120** and a first collimator lens **130-1** (e.g., a first aspheric lens (A-lens)). The optical assembly may further include a pump chip (or pump laser) **160** to provide pump light that has a first wavelength (e.g., 980 nm), which may couple into the output leg through a second collimator lens **130-2** (e.g., a second A-lens), a WDM filter **140**, the first collimator lens **130-1**, and a compensator **150** arranged between

the dual fiber pigtail **120** and the first collimator lens **130-1**.

(10) As shown in FIGS. **1A-1C**, the optical assembly may be a pump combiner, a forward pump EDFA, or another suitable device in which the single stage isolator core **110** is provided in the input leg to receive signal light having a second wavelength (e.g., 1550 nm), which is combined in the output leg with the pump light from the pump chip **160**. In this case, the pump light may couple into the output leg through the second collimator lens **130-2**, the WDM filter **140** (which is transmissive at the wavelength of the pump light), the first collimator lens **130-1**, and the compensator **150**. Alternatively, in a reverse pump EDFA (not shown) or another suitable device in which the pump light is to be coupled into the input leg, the single stage isolator core **110** may be provided in the output leg, and the pump light may be coupled into the input leg through the second collimator lens **130-2**, the WDM filter **140**, the first collimator lens **130-1**, and the compensator **150**. Accordingly, the isolator core **110** and the compensator **150** may be arranged adjacent to one another in the input leg and the output leg (or vice versa), between the dual fiber pigtail **120** and the first collimator lens **130-1**.

(11) In some implementations, the optical assembly may be provided in an integrated module having a highly-integrated pump platform (HIPP) design. For example, as shown in FIG. **1A**, the integrated module may include various mechanical components, such as metal holders **170** to fix or otherwise support optical components such as the dual fiber pigtail **120**, the isolator core, the compensator **150**, the collimator lenses **130**, and the WDM filter **140**. Additionally, in cases where the isolator core **110** is integrated with the pump chip **160**, the integrated module may include a package **180** to mount the pump chip **160** and package pins **190** to interface with one or more external components. In another example, the isolator core **110** may be used in the input leg and/or the output leg of a free space optics configuration or another suitable configuration that may be part of a larger assembly without having optical fibers that directly couple into and/or out of the isolator core **110** and the compensator **150**. Accordingly, it will be appreciated that the assemblies described herein are exemplary only, and that the isolator core **110** and/or variants thereof may be used in any suitable optical device or application.

(12) In some implementations, by positioning the isolator core **110** and the compensator **150** between the dual fiber pigtail **120** and the first collimator lens **130-1**, an optical isolation of more than 18 decibels (dB) may be achieved between the input leg and the output leg. In some implementations, the compensator **150** may compensate an optical path between the input leg and the output leg. In some implementations, in the optical assembly shown in FIG. **1A**, a fiber core of the input leg and a fiber core of the output leg may have a fiber-core-to-fiber-core distance of 460 microns (μm), the collimator lenses **130** may have a 2.7 millimeter (mm) focal length, and the WDM filter **140** may be a 980 nm (transmissive)/1550 nm (reflective) filter. The input leg and the output leg may be a thermal expanded core (TEC) fiber that expands a fiber mode diameter from 6 μm to 9 μm . In this way, an optical device including the isolator core **110** may have a diameter of 3.0 mm or less.

(13) In some implementations, as described herein, the isolator core **110** may be configured to receive and provide a lateral displacement walk-off to transmit the signal light towards the first collimator lens **130-1**, which produces a collimated light beam that is reflected by the WDM filter **140** (e.g., the WDM filter **140** is transmissive at the wavelength associated with the pump light and reflective at the wavelength associated with the signal light). In this way, the pump light and the signal light are combined and couple into the compensator **150** through the first collimator lens **130-1**. Furthermore, by positioning the isolator core **110** between the dual fiber pigtail **120** and the first collimator lens **130-1** (rather than between the first collimator lens **130-1** and the WDM filter **140**), the isolator core **110** does not block the pump light traveling in the reverse direction relative to the (input) signal light. In some implementations, to laterally displace the signal light, the isolator core **110** may include a set of layers or components that are designed to spatially separate and combine orthogonal polarization components of the signal light with a low PMD, a low PDL,

and/or a low IL. As described herein, the isolator core **110** may include a Faraday rotator, an input polarizer (e.g., a first birefringent crystal plate), an output polarizer (e.g., a second birefringent crystal plate), and an additional component (e.g., a third birefringent crystal plate or a half-wave plate) that are bonded together (e.g., by an adhesive material, such as epoxy and/or the like). Furthermore, as described herein, a thickness and an angle of the various layers or components of the isolator core **110** may be controlled to create the lateral displacement walk-off.

(14) For example, as shown in FIG. 1D, the isolator core **110** may include a Faraday rotator **112** made from garnet, and a first birefringent crystal plate **114-1** to receive input light that includes orthogonal polarization components (e.g., O-light and E-light, which may respectively refer to light having a horizontal polarization and light having a vertical polarization). In some implementations, the input light may be spatially separated into the orthogonal polarization components by the first birefringent crystal plate **114-1**. For example, the first birefringent crystal plate **114-1** may include a birefringent material, such as Yttrium orthovanadate (YVO.sub.4), or another suitable material that has a refractive index that depends on the polarization and propagation direction of light.

Accordingly, because the orthogonal polarization components have different polarizations, the first birefringent crystal plate **114-1** has a different refractive index with respect to the O-light and the E-light, which causes the O-light and the E-light to be spatially separated. However, at the output of the isolator core **110**, the O-light and the E-light has to combine together exactly to avoid polarization-dependent loss or other performance issues. Accordingly, after the O-light and the E-light passes through the first birefringent crystal plate **114-1**, the O-light and the E-light are rotated by the Faraday rotator **112**, and another pair of birefringent crystal plates **114-2**, **114-3** are provided to combine the O-light and the E-light.

(15) In general, controlling the thickness and angle of the Faraday rotator **112** and the three birefringent crystal plates **114** may be challenging during an assembly process, as the Faraday rotator **112** and the three birefringent crystal plates **114** may have different material tolerances (e.g., plus or minus ten microns relative to a desired thickness) and different assembly tolerances (e.g., plus or minus a number of degrees relative to a desired angle). Accordingly, in some implementations, the Faraday rotator **112** may have a fixed orientation with respect to one of the adjacent birefringent crystal plates **114** (e.g., birefringent crystal plate **114-1** or birefringent crystal plate **114-2**), and the other two birefringent crystal plates **114** may be rotated with two degrees of freedom to tune the necessary angles to ensure that the O-light and the E-light combines precisely at the output from the third birefringent plate **114-3**. For example, in some implementations, the Faraday rotator **112** and the first birefringent crystal plate **114-1** may be fixed, and the other birefringent crystal plates **114-2**, **114-3** may be rotated to compensate for the different material and/or assembly tolerances. Alternatively, the Faraday rotator **112** and the second birefringent crystal plate **114-2** may be fixed, and the other birefringent crystal plates **114-1**, **114-3** may be rotated to compensate for the material and/or assembly tolerances. In this way, as shown in FIG. 1D, the O-light and the E-light may be combined in the light that is output from or otherwise transmitted by the isolator core **110**. In one example, Table 1 below shows design characteristics of the isolator core **110** that may achieve a lateral displacement walk-off, where α , β , and γ are azimuth angles of the optical axes of each birefringent crystal plate **114**, one of which is fixed with respect to the Faraday rotator **112** and two of which are rotatable.

(16) TABLE-US-00001

TABLE 1	Thickness (mm)	$\cos(\alpha)$	$\cos(\beta)$	$\cos(\gamma)$	YVO.sub.4-1	0.450	0.688	0.000	0.726	Garnet	0.42	-45° rotation	YVO.sub.4-2	0.300	-0.465	0.507	0.726	YVO.sub.4-3	0.300	0.507	0.465	0.726

(17) Alternatively, in some implementations, the single stage isolator core **110** may have a design as shown in FIG. 1E, where the isolator includes a first birefringent crystal plate **114-1**, a half-wave plate (HWP), a Faraday rotator **112**, and a second birefringent crystal plate **114-2**. In this case, different layers (e.g., the first birefringent crystal plate **114-1** and the second birefringent crystal plate **114-2**) may be diced from one larger piece (e.g., of YVO.sub.4). Accordingly, whereas the

design shown in FIG. 1D achieves precise control over the thickness and angle of the components of the isolator core **110** by rotating two of three birefringent crystal plates **114**, the design shown in FIG. 1E may ensure that the O-light and the E-light combine together to minimize PDL and/or insertion loss because the different layers have identical material tolerances. In this way, even if the layers in the isolator core **110** are not tuned perfectly, the O-light and the E-light will combine at the output from the isolator core **110** because the birefringent crystal plates **114-1**, **114-2** layers have identical thicknesses and respective magnetic orientations (e.g., reversed or identical magnetic orientations) to cancel a material tolerance and an assembly tolerance with respect to one another.

(18) In some implementations, as described above, the Faraday rotator **112** may be made from garnet. For example, the Faraday rotator **112** may be made from a rare-earth iron garnet (RIG) single crystal that generally needs a magnetic field to work as the Faraday rotator **112** or and/or a magnet-free garnet (referred to herein as a latching garnet). For example, when the Faraday rotator **112** is made from RIG single crystal and the RIG is exposed to an external magnetic field that is greater than a saturation field, a magnetic domain in a direction opposite to the external magnetic field is neutralized and a single magnetic domain is formed in the direction in which the direction of magnetization is aligned. In this way, because light passing through the RIG with a single magnetic domain is subject to Faraday rotation in one direction, the RIG single crystal can be used as the Faraday rotator **112**. Alternatively, in cases where the Faraday rotator **112** is made from latching (or magnet-free) garnet, the Faraday rotator **112** may retain its single domain structure once magnetized in a strong external magnetic field even if the external magnetic field is later removed (although the single domain structure can be reversed by applying a strong external magnetic field). Accordingly, the latching garnet requires no external magnetic field when used as the Faraday rotator **112**. In other words, the latching garnet functions as a Faraday rotator **112** without being exposed to an external magnetic field.

(19) As indicated above, FIGS. 1A-1E are provided as one or more examples. Other examples may differ from what is described with regard to FIGS. 1A-1E.

(20) FIGS. 2A-2C illustrate one or more example implementations **200** of a dual stage isolator including two isolator cores that are arranged in parallel. For example, as shown in FIGS. 2A-2C, the dual stage isolator may include a first isolator core **210-1** and a second isolator core **210-2** arranged in parallel. For example, as shown in FIG. 2A, the first isolator core **210-1** may be provided in an input leg of an optical assembly and the second isolator core **210-2** may be provided in an output leg of the optical assembly. In this case, the dual stage isolator may be used with a collimator lens **230** and a WDM filter **240** in a standalone isolator application to only transmit light having a particular wavelength (e.g., 1550 nm), without integration with a pump.

(21) As shown in FIG. 2B, the first isolator core **210-1** and the second isolator core **210-2**, collectively referred to as dual stage isolator **210**, includes a first set of layers and a second set of layers that are diced from the same pieces. For example, the first isolator core **210-1** includes a first HWP **216-1** and a first Faraday rotator **212-1** arranged between a first pair of birefringent crystal plates **214-11**, **214-12**, and the second isolator core **210-2** includes a second HWP **216-2** and a second Faraday rotator **212-2** arranged between a second pair of birefringent crystal plates **214-22**, **214-21**. In this case, the first isolator core **210-1** and the second isolator core **210-2** each have four crystal layers, the first isolator core **210-1** and the second isolator core **210-2** are both diced from one large piece, and the first isolator core **210-1** and the second isolator core **210-2** are arranged adjacent to one another with crystal layers in different orientations. For example, in some implementations, the first Faraday rotator **212-1** and the second Faraday rotator **212-2** may be oriented such that a magnetic domain direction of the first Faraday rotator **212-1** is reversed relative to the magnetic domain direction of the second Faraday rotator **212-2**. Accordingly, because the various layers are diced from the same raw material, the dual stage isolator **210** may include two isolator cores **210-1**, **210-2** in which the respective layers have certain properties (e.g., reversed orientations, identical thicknesses, identical angles, and/or the like) that cause the isolator

cores **210-1**, **210-2** to cancel material and/or assembly tolerances of each other. In this way, input light received at the input polarizer of the first isolator core **210-1** (e.g., birefringent plate **214-11**) may be spatially separated into components (e.g., O-light and E-light) having orthogonal polarizations, and the components may recombine at the output polarizer of the first isolator core **210-1** (e.g., birefringent plate **214-12**) even if the first set of layers is not perfectly tuned. In a similar respect, after the light that is laterally displaced by the first isolator core **210-1** passes through the collimator lens **230** and is reflected back towards the output leg by the WDM filter **240**, the reflected light received at the input polarizer of the second isolator core **210-2** (e.g., birefringent plate **214-21**) may be spatially separated into the orthogonal components, which again recombine at the output polarizer of the second isolator core **210-2** (e.g., birefringent plate **214-22**) such that the second isolator core **210-2** laterally displaces the light received at the input polarizer. Furthermore, as shown in FIG. 2C, and by reference number **250**, any backward travelling light that is reflected from the output leg to the input leg is separated into the components that have the orthogonal polarizations. In this way, the design of the dual stage isolator **210** causes the backward travelling light to be directed or otherwise steered away from a spot or position where the light is input through the input leg. In this way, the backward travelling light is not coupled into the input leg, which achieves high isolation between the input leg and the output leg.

(22) As indicated above, FIGS. 2A-2C are provided as one or more examples. Other examples may differ from what is described with regard to FIGS. 2A-2C.

(23) FIGS. 3A-3C illustrate one or more example implementations **300** of a dual stage isolator including two isolator cores that are arranged in series. For example, as shown in FIGS. 3A-3C, the dual stage isolator may include a first isolator core **310-1** and a second isolator core **310-2** arranged in series. For example, as shown in FIG. 3A, the first isolator core **310-1** and the second isolator core **310-2** may both be provided in an input leg of an optical assembly with a compensator **350** provided in the output leg. Alternatively, in some implementations, the first isolator core **310-1** and the second isolator core **310-2** (collectively dual self-compensated isolator core **310**) may both be provided in the output leg of an optical assembly with the compensator **350** provided in the input leg. In such cases, the dual stage isolator may be used with a collimator lens **330** and a WDM filter **340** in a standalone isolator application to only transmit light having a particular wavelength (e.g., 1550 nm), without integration with a pump, or the dual stage isolator may be used in a pump combiner or another suitable device or application to achieve a higher isolation relative to the single stage design(s) described above.

(24) As shown in FIG. 3B, the first isolator core **310-1** and the second isolator core **310-2** respectively include a first set of layers and a second set of layers that are diced from the same pieces. For example, the first isolator core **310-1** includes a first HWP **316-1** and a first Faraday rotator **312-1** arranged between a first pair of birefringent crystal plates **314-11**, **314-12**, and the second isolator core **310-2** includes a second HWP **316-2** and a second Faraday rotator **312-2** arranged between a second pair of birefringent crystal plates **314-22**, **314-21**. In this case, where the first isolator core **310-1** and the second isolator core **310-2** are arranged in series, the first Faraday rotator **312-1** and the second Faraday rotator **312-2** may be oriented such that a magnetic domain direction of the first Faraday rotator **312-1** is the same as the magnetic domain direction of the second Faraday rotator **312-2**. Accordingly, because the various layers are diced from the same raw material (e.g., the same garnet and/or YVO.sub.4 pieces), the dual stage isolator **310** may include two isolator cores **310-1**, **310-2** that have certain properties (e.g., identical magnetic orientations, thicknesses, angles, and/or the like). In this way, input light received at the input polarizer of the first isolator core **310-1** (e.g., birefringent plate **314-11**) may be spatially separated into components (e.g., O-light and E-light) having orthogonal polarizations, and the components may recombine at the output polarizer of the first isolator core **310-1** (e.g., birefringent plate **314-12**) even if the first set of layers is not perfectly tuned. In a similar respect, the light that is laterally displaced by the first isolator core **310-1** is received at the input polarizer of the second isolator

core **310-2** (e.g., birefringent plate **314-21**), where the light is again spatially separated into orthogonal components that recombine at the output polarizer of the second isolator core **310-2** (e.g., birefringent plate **314-22**). Accordingly, after the light laterally displaced by the second isolator core **310-2** passes through the collimator lens **330**, the light that includes the components with orthogonal polarizations is reflected back towards the output leg by the WDM filter **340**. Furthermore, as shown in FIG. **3C**, and by reference number **350**, any backward travelling light that is reflected from the output leg to the input leg is separated into the components that have the orthogonal polarizations. In this way, the design of the dual stage isolator **310** causes the backward travelling light to be directed or otherwise steered away from a spot or position where the light is input through the input leg, whereby the backward travelling light is not coupled into the input leg and high isolation is achieved between the input leg and the output leg.

(25) As indicated above, FIGS. **3A-3C** are provided as one or more examples. Other examples may differ from what is described with regard to FIGS. **3A-3C**.

(26) The foregoing disclosure provides illustration and description, but is not intended to be exhaustive or to limit the implementations to the precise form disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the implementations. Furthermore, any of the implementations described herein may be combined unless the foregoing disclosure expressly provides a reason that one or more implementations may not be combined.

(27) As used herein, the terms circuit, integrated circuit, chip, chipset, die, semiconductor device, electronic device, and/or the like are intended to be broadly construed as applicable to the various implementations described herein, as these terms can be used interchangeably in the field of electronics. With respect to a circuit, an integrated circuit, and/or the like, power, ground, and various signals may be coupled between and among circuit elements (e.g., resistors, inductors, capacitors, transistors, and/or the like) via physical, electrically conductive connections. Such a point of connection may be referred to as an input, output, input/output (I/O), terminal, line, pin, pad, port, interface, or similar variants and combinations. Although connections between and among circuits can be made by way of electrical conductors, circuits and other circuit elements may additionally, or alternatively, be coupled by way of optical, mechanical, magnetic, electrostatic, electromagnetic, and/or other suitable interfaces.

(28) It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware, software, circuitry, or a combination thereof. The actual specialized control hardware, software code, or circuitry used to implement these systems and/or methods is not limiting of the implementations. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code—it being understood that software and hardware (e.g., integrated circuits) can be designed to implement the systems and/or methods based on the description herein.

(29) Although particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one claim, the disclosure of various implementations includes each dependent claim in combination with every other claim in the claim set.

(30) No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items, and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the term “set” is intended to include one or more items (e.g., related items, unrelated items, a combination of related and unrelated items, etc.), and may be used interchangeably with “one or

more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

(31) Further, spatially relative terms, such as “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the apparatus, device, and/or element in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

Claims

1. A device, comprising: a first plurality of birefringent crystal plates comprising: a first birefringent crystal plate to separate input light; a second birefringent crystal plate to combine light; wherein the first birefringent crystal plate and the second birefringent crystal plate are rotatable to respective angles that cause the light to be combined into a first output light that is laterally displaced; and a third birefringent crystal plate configured with a fixed orientation with respect to a first Faraday rotator configured to be between the first birefringent crystal plate and the second birefringent crystal plate; and a second plurality of birefringent crystal plates, coupled to the first plurality of birefringent crystal plates in series, comprising: a fourth birefringent crystal plate to separate the first output light; a fifth birefringent crystal plate to combine the light separated by the fourth birefringent crystal plate, wherein the fourth birefringent crystal plate and the fifth birefringent crystal plate are rotatable to respective angles that cause the light to be combined into a second output light that is laterally displaced; and a sixth birefringent crystal plate configured with a fixed orientation with respect to a second Faraday rotator.
2. The device of claim 1, wherein the third birefringent crystal plate is a half-wave plate.
3. The device of claim 1, wherein the third birefringent crystal plate is configured to be between the first Faraday rotator and the first birefringent crystal plate.
4. The device of claim 1, wherein the device is configured to be arranged at an input leg of an optical assembly or an output leg of the optical assembly.
5. The device of claim 1, wherein the third birefringent crystal plate and the sixth birefringent crystal plate are half-wave plates.
6. The device of claim 1, wherein the first Faraday rotator and the second Faraday rotator are configured to have a same magnetic domain direction.
7. The device of claim 1, wherein the first output light is received as input by the second plurality of birefringent crystal plates.
8. The device of claim 1, wherein the second plurality of birefringent crystal plates are configured to be coupled to a collimator lens.
9. An optical assembly, comprising: a collimator lens; a first plurality of birefringent crystal plates comprising: a first birefringent crystal plate to separate input light, a second birefringent crystal plate to combine light, wherein the first birefringent crystal plate and the second birefringent crystal plate are rotatable to respective angles that cause the light to be combined into a first output light that is laterally displaced; and a third birefringent crystal plate configured with a fixed orientation with respect to a first Faraday rotator configured to be between the first birefringent crystal plate and the second birefringent crystal plate; and a second plurality of birefringent crystal plates, coupled to the first plurality of birefringent crystal plates in series, comprising: a fourth

birefringent crystal plate to separate the first output light; a fifth birefringent crystal plate to combine the light separated by the fourth birefringent crystal plate, wherein the fourth birefringent crystal plate and the fifth birefringent crystal plate are rotatable to respective angles that cause the light to be combined into a second output light that is laterally displaced; and a sixth birefringent crystal plate configured with a fixed orientation with respect to a second Faraday rotator.

10. The optical assembly of claim 9, wherein the third birefringent crystal plate is a half-wave plate.

11. The optical assembly of claim 9, wherein the third birefringent crystal plate is configured to be between the first Faraday rotator and the first birefringent crystal plate.

12. The optical assembly of claim 9, wherein the second plurality of birefringent crystal plates is configured to be arranged at an input leg of an optical assembly or an output leg of the optical assembly.

13. The optical assembly of claim 9, wherein the third birefringent crystal plate and the sixth birefringent crystal plate are half-wave plates.

14. The optical assembly of claim 9, wherein the first Faraday rotator and the second Faraday rotator are configured to have a same magnetic domain direction.

15. The optical assembly of claim 9, wherein the first output light is received as input by the second plurality of birefringent crystal plates.

16. The optical assembly of claim 9, wherein the second plurality of birefringent crystal plates are configured to be coupled to the collimator lens.
